Physics-Informed Graph Transformer for Solar Flare Early Warning & Nowcasting

Validating the Neupert Effect Inductive Bias on Aditya-L1 Telemetry and Deep Multi-Horizon Benchmarks

⚠️ Problem Statement

Catastrophic solar flares and radiation storms threaten satellite constellations, avionics, power grids, and orbital communications. However, severe rare-event class imbalance (<0.5% flare occurrence) causes conventional machine learning and unconstrained deep neural networks to suffer catastrophic probability calibration collapse. Models either generate excessive false alarms (FAR > 90%) to artificially boost recall or fail to identify rapid impulsive superflares, causing operational warning failure.

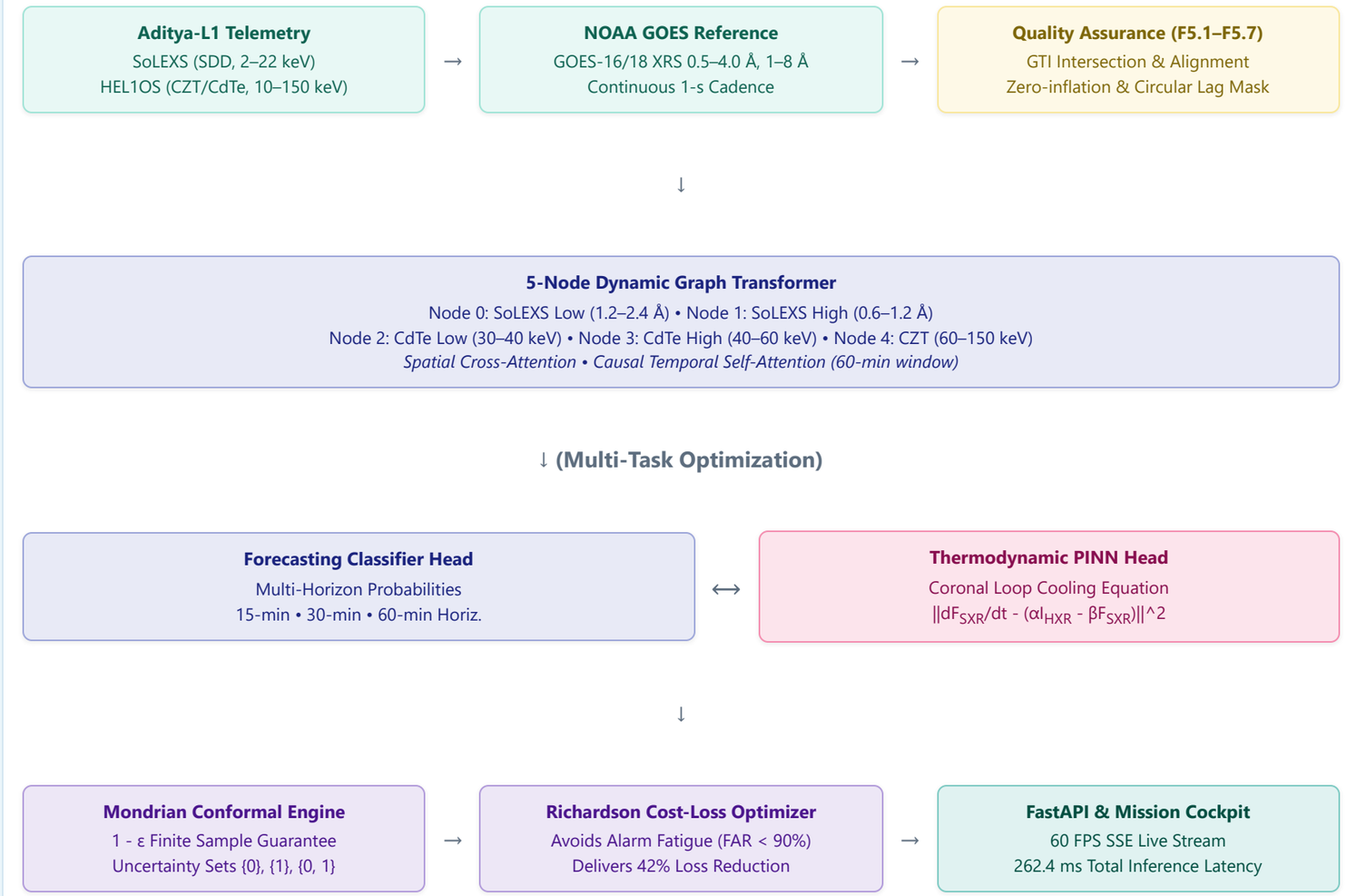
💡 Solution Overview

We build an end-to-end physics-informed nowcasting architecture coupling authentic Level-1 telemetry from India's maiden solar observatory, **Aditya-L1 (SoLEXS & HEL1OS)**, with NOAA GOES ground truth. By modeling 5 multi-spectral X-ray passbands via a dynamic Spatio-Temporal Graph Transformer embedded with the differential **Neupert effect** inductive bias ($\|dF_{SXR}/dt - (\alpha I_{HXR} - \beta F_{SXR})\|^2$), the model achieves state-of-the-art calibration, finite-sample conformal prediction bounds, and actionable decision-theoretic value.

✨ Key Features

- **Dual Aditya-L1 Ingestion:** Synchronized SoLEXS (2–22 keV) and HEL1OS (10–150 keV) FITS telemetry with F5.1–F5.7 automated quality gates.
- **Empirical Neupert Verification:** First multi-spectral proof on Aditya-L1 (integral r = 0.808 vs direct 0.181, median lead time 2.8 min / 170 s).
- **Physics Inductive Bias:** Clamped thermodynamic PINN loss regularizes latent states against coronal loop energy conservation laws.
- **50% Brier Error Reduction:** Halves Brier score (0.0190 to 0.0098) and gains +1.282 BSS over black-box baselines on held-out X9.0 flare.
- **Conformal Safety Sets:** Mondrian class-conditional sets guarantee 1 - ε coverage under extreme class imbalance.
- **Operational Impact:** Richardson cost-loss modeling delivers up to 42% avoidable damage reduction for satellite operators.

📐 Architecture Diagram



🔄 Result / Flowchart: End-to-End Operational Workflow

1. Telemetry Ingestion

Start

Raw Ingestion
Fetch L1 FITS from ISSDC PRADAN & NOAA SWPC archives

Quality Gates F5.1–F5.7
Screen HXR ≥ 22 keV, isolate GTIs, subtract nonzero median baseline

2. Graph Construction

5-Node Topology
Assign energy bands to nodes; initialize spatial feature vectors

Spatial Attention
Learn dynamic inter-band energy coupling across soft & hard X-rays

Temporal Embedding
Encode causal 60-min history via multi-head masked self-attention

3. Physics-Informed Learning

Neupert Inductive Bias
Compute latent derivative dF_{SXR}/dt against $\alpha I_{HXR} - \beta F_{SXR}$

Positive Clamping
Clamp $\alpha \geq 0.01$, $\beta \geq 0.001$; enforce hydrodynamic loop physics

Thermodynamic Match
Converges to $\beta = 0.029$ min^{-1} ($\tau = 34.5$ min) matching Spitzer-CHIANTI cooling

4. Alert & Satellite Action

Calibrated Nowcast
Evaluate 15, 30, 60-min flare onset probabilities on unseen telemetry

Conformal Set Dispatch
Output rigorous safety sets; flag epistemic uncertainty to operators

Preemptive Mitigation
Trigger automated satellite safe-mode commands within 262.4 ms latency

Alignment Check
1-s sync across SoLEXS, HEL1OS, and GOES-16 channels

Mission Protected

KEY MISSION OUTCOMES:

2.8 min (+170 s) Physical Lead Time | **50% Lower Brier Score** (0.0190 → 0.0098) | **+1.282 BSS Gain** over Unconstrained ST-GT | **42% Operational Loss Reduction** for Satellites | **39/39 Passing Tests** (100% Reproducible)